

Annie Prasanna Manoharan - Software Engineer

+1(479)301-1366 | anniepr0501@gmail.com | <https://annie0159.github.io/Portfolio/> | <https://github.com/Annie0159>

EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Cryptography, Privacy Enhancing Techniques

TECHNICAL SKILLS

Languages: Python, C++, C#, C, Java, SQL, HTML, CSS

Databases: SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, ClickHouse

Software development: Object-Oriented Programming, Data Structures & Algorithms, Software Design Patterns, Multithreading & Concurrency, Memory Management, Troubleshooting

Testing & Quality: Test Automation, Unit Testing, Test Case Design, Software Validation, Integration Testing, Debugging, Test Coverage Optimization

Tools & Practices: Git, Bitbucket, Jira, Agile, Scrum, SDLC, Version Control, Code Review, Technical Documentation

PROFESSIONAL EXPERIENCE

Teaching Assistant, Department of Computer Science and Engineering, University of Arkansas

Sep 2024 – May 2026

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals
- Assisted students with control flow, functions, arrays, programming logic, debugging, and problem-solving techniques
- Supported Database Management Systems instruction for approximately 180 students, assisting with database concepts, assignments, assessments, and technical questions

Research Assistant, AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

- Developed automated data-processing and annotation pipelines that used previously annotated data to automatically generate bounding boxes, keypoints, and structured datasets from raw image and video frames, reducing repetitive manual processing
- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Automated image-processing workflows for mask generation, image alignment, and inpainting, reducing manual preparation required for high-resolution image datasets

Project Engineer, Soliton Technologies, India

July 2022 – June 2024

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Implemented reusable software components and automated workflows within an object-oriented application architecture, improving development and testing efficiency
- Developed customized software solutions for Medtronic as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements
- Integrated automated testing and validation workflows to improve test coverage, reduce manual execution, and support repeatable software verification
- Collaborated with cross-functional engineering teams to debug system-level issues, troubleshoot application behavior, optimize workflows, and deliver production-ready software

PROJECTS

Expense Tracker with Predictive Analytics

C++ | MySQL | Python | SQL | Database Integration

- Designed and implemented a cross-language expense management application integrating C++ business logic with a MySQL database for storing, retrieving, categorizing, and processing historical expense data
- Developed and optimized SQL queries to generate category-based expense summaries and reports, integrating database operations with the application's business logic
- Integrated Python-based analytics with the application workflow to analyze historical spending patterns and generate next-month expense predictions

Pacman Search Algorithms

Python | Data Structures & Algorithms | BFS | A* | Heuristics

- Designed and implemented search algorithms in Python, defining custom problem states, successor functions, goal conditions, and state representations for maze-based environments
- Implemented Breadth-First Search (BFS) and A* search with admissible heuristics to efficiently navigate complex mazes while minimizing path cost
- Developed a greedy approximation strategy for large state spaces where optimal search was computationally expensive, improving scalability for complex search problems